

This Zenodo deposit contains a publicly available description of the Dataset:

"Single-cell RNAseq of lamina propria in wild type and LRRK2 G2019S mice after *C. rodentium* infection".

Dataset Description:

We performed 10X Genomics single-cell RNAsequencing of colonic lamina propria cells from wild type and LRRK2 G2019S mice following 1-week post *C. rodentium* infection. The cells were pooled from 3 mice per group of both sexes at 8-12 weeks of age. This dataset contains raw FASTQ files from mouse colonic lamina propria, sorted into 4 groups namely wild type and LRRK2 G2019S uninfected and infected mice. Sequencing was performed using NovaSeq 6000 S4 PE 100bp. Reads were processed using the 10X Genomics Cell Ranger Single Cell 2.0.0 pipeline. FASTQs generated from sequencing output were aligned to the mouse GRCh38 reference genome using STAR algorithm 2.7.3a.

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ASAP Team: Desjardins

Dataset Name: desjardins-mouse-sc-rnaseq-colon-immune-lrrk2, v1.0

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ASAP Lab:

ASAP Project: LRRK2 G2019S mutation incites increased cell-intrinsic neutrophil effector functions and intestinal inflammation in a model of infectious colitis

Project Description: Parkinson's Disease (PD) is a neurodegenerative disorder often preceded by gastrointestinal dysfunction. Mutations in leucine-rich repeat kinase 2 (LRRK2) are known risk factors for both PD and inflammatory bowel disease (IBD), suggesting a link between PD and the gastrointestinal tract. Using single-cell RNA-sequencing and spectral flow cytometry, we demonstrated that the *Lrrk2* Gly2019Ser (G2019S) mutation is associated with an increased neutrophil presence in the colonic lamina propria during *Citrobacter rodentium* infection. This concurred with a Th17 skewing, upregulated *Il17a*, and greater colonic pathology during infection. In vitro experiments showed enhanced kinase-dependent neutrophil chemotaxis and neutrophil extracellular trap (NET) formation in *Lrrk2* G2019S mice compared to wild-type counterparts. Our results add to the understanding of LRRK2-driven immune cell dysregulation and its contribution to PD, offering insights into potential biomarkers for early diagnosis and intervention in PD.

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This dataset is made available to researchers via the ASAP CRN Cloud: cloud.parkinsonsroadmap.org. Instructions for how to request access can be found in the [User Manual](#).

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This Zenodo deposit was created by the ASAP CRN Cloud staff on behalf of the dataset authors. It provides a citable reference for a CRN Cloud Dataset